

THANATHAI LERTPETCHPUN

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RESEARCH INTEREST

Controllable Speech Generation, Speech Emotion Recognition, Speaker Verification, Voice Anti-spoofing

EDUCATION

PhD in Electrical and Computer Engineering, University of Southern California 2024 - Present

Advisor: **Prof. Shrikanth Narayanan**

Current GPA: 3.83/4.00

Bachelor in Computer Engineering, Chulalongkorn University 2018 - 2022

Thesis Advisor: **Prof. Ekapol Chuangsuwanich**

GPA: 3.82/4.00, CS-only 4.00/4.00

WORK EXPERIENCE

Graduate Research Assistant 08/2025 - Present

Signal Analysis and Interpretation Labortory (SAIL) *Los Angeles, CA*

- Conduct research in speech emotion recognition, text to speech, accent conversion, and voice conversion.
- Develop a framework to evaluate privacy for identity and attribute in speech.

Undergrad Research and Teaching Assistant 05/2021 - 08/2024

Chulalongkorn University *Bangkok, Thailand*

- Taught undergraduate students Computer Programming, Discrete Mathematics, Linear Algebra, Algorithm Design, and Pattern Recognition.
- Created and graded coursework for Computer Programming, Linear Algebra and Pattern Recognition.

Data Scientist 06/2021 - 07/2021

Tencent (Thailand) *Bangkok, Thailand*

- Created a full machine learning pipeline for news recommendation, including text scraping, training model, inferencing model, setting up multiple GPUs for training and storing data.
- Designed and implemented a complete architecture for applying machine learning in music recommendation including scraping data, cleaning data, fine-tuning model, and inferencing model.

PROJECTS

Anonymous Real Time Speech (ARTS) (USC - JHU - Meaning Company) 10/2024 - Present

- Developing zero-shot framework for multilingual TTS on British, Spanish, and Indian Accent.
- Analyzing the effect of speaker embedding on accent strength using phonemes rules to control the effect. (1 ICASSP2026 publication)
- Proposed frameworks to develop state-of-the-art speech emotion recognition (SER) for predicting both emotional attributes and categorical emotions. They achieved 1st rank in emotional attributes and 2nd rank in categorical attributes in Naturalistic Conditions Challenge (2 Interspeech2025 publications)

Speaker Recognition System and Anti-spoofing System. (Chulalongkorn University) 08/2022 - 05/2024

- Proposed an approach to amplify artifacts in spoofed utterances using speech enhancement and show its robustness on different countermeasures and speech enhancement models (1 Interspeech2025 publication).
- Proposed a new normalization layer to address the problem of mismatched emotions and languages in speaker verification. (1 Interspeech2023 publication).

SELECTED PUBLICATIONS

(ICASSP, 2026) **Thanathai Lertpetchpun***, Yoonjeong Lee*, Thanapat Trachu, Jihwan Lee, Tiantian Feng, Dani Byrd, Shrikanth Narayanan. “Quantifying Speaker Embedding Phonological Rule Interactions in Accented Speech Synthesis”

(Interspeech, 2025) **Thanathai Lertpetchpun***, Tiantian Feng*, Dani Byrd, Shrikanth Narayanan. “Developing a High-performance Framework for Speech Emotion Recognition in Naturalistic Conditions Challenge for Emotional Attribute Prediction.”

(Interspeech, 2025) Tiantian Feng*, **Thanathai Lertpetchpun***, Dani Byrd, Shrikanth Narayanan. “Developing A Top-tier Framework in Naturalistic Conditions Challenge for Categorized Emotion Prediction: From Speech Foundation Models and Learning Objective to Data Augmentation and Engineering Choices.”

(Interspeech, 2025) Thanapat Trachu*, **Thanathai Lertpetchpun***, Ekapol Chuangsuwanich. “Amplifying Artifacts with Speech Enhancement in Voice Anti-spoofing.”

(Interspeech, 2023) **Thanathai Lertpetchpun**, Ekapol Chuangsuwanich. “Instance-based Temporal Normalization for Speaker Verification.”

(* denotes equal contribution)

AWARDS AND HONORS

ICASSP 2026 Travel Grant Student Travel Grant for ICASSP 2026

SER in Naturalistic Conditions Challenge (2025) Rank 1st in Attribute Prediction

SER in Naturalistic Conditions Challenge (2025) Rank 2nd in Categorical Prediction

Viterbi School of Engineering Fellowship (2024) Top-off fellowship selected among incoming Ph.D. students

Interspeech 2023 Travel Grant Student Travel Grant for Interspeech2023

Thailand Olympiad Informatics (2017) Ranked in the top-15 among all Thai high school contestants

SKILLS

Languages: Thai (Native), English (Fluent).

Computer Languages: Python, C/C++, Bash Script, JavaScript, Node.js.

Framework: Pytorch, Espnet, Huggingface, Speechbrain.

SERVICE

Social Chair at SAIL (2026 - Present)

Deputy Social Chair at SAIL (2025)

Military Service (2015-2018)